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|---------------------|-----------------------------------|
| <b>Started on</b>   | Sunday, 1 February 2026, 10:08 AM |
| <b>State</b>        | Finished                          |
| <b>Completed on</b> | Sunday, 1 February 2026, 11:00 AM |
| <b>Time taken</b>   | 51 mins 34 secs                   |
| <b>Grade</b>        | Not yet graded                    |

**Question 1**

Correct

Mark 0.50 out of 0.50

SSR measures how far the predicted values are from the mean of the actual values.

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

**Question 2**

Correct

Mark 0.50 out of 0.50

Min-max normalization is more robust to outliers than z-score normalization.

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

**Question 3**

Correct

Mark 0.50 out of 0.50

Which of the following is TRUE about KNN?

- ☐ a. It assumes linear decision boundaries
- ☐ b. It requires gradient descent
- ☐ c. It builds an explicit model during training
- ☒ d. It stores all training data ✓

The correct answer is: It stores all training data

**Question 4**

Incorrect

Mark 0.00 out of 0.50

A model predicts whether a customer will spend more than \$500 next month. What is the ML task?

- ☐ a. Classification
- ☒ b. Regression ✗
- ☐ c. Anomaly Detection
- ☐ d. Clutsering

The correct answer is: Classification

**Question 5**

Incorrect

Mark 0.00 out of 0.50

A dataset has labels, but the goal is to discover hidden patterns, not predict the labels. What is the correct task?

- ☐ a. Regression
- ☐ b. Classification
- ☒ c. Association Rule ✗
- ☐ d. Clustering

The correct answer is: Clustering

**Question 6**

Correct

Mark 0.50 out of 0.50

Sentiment analysis with labels {positive, neutral, negative} is a multi-class problem.

☒ True ✓☐ False

The correct answer is 'True'.

**Question 7**

Incorrect

Mark 0.00 out of 0.50

Logistic regression minimizes MSE.

☒ True ✗☐ False

The correct answer is 'False'.

**Question 8**

Complete

Marked out of 2.00

A logistic regression model is used to predict whether a person has a certain disease.

The model is defined as:  $z = b_1x_1 + b_2x_2 + a$

where:

- $x_1$  represents **Age**
- $x_2$  represents **Smoking status** (1 = Smoker, 0 = Non-smoker)

Given the following values:

- $b_1 = 0.5$
- $b_2 = 1.2$
- $a = -25$
- Age = 40
- Smoker = 1

Using a threshold of **0.5**, determine whether the person is classified as **diseased or not diseased**.

$$z = b_1x_1 + b_2x_2 + a = -3.8$$

$$g(z) = 1 / (1 + e^{-z}) = 0.02$$

the person is classified as diseased

**Question 9**

Correct

Mark 0.50 out of 0.50

Dummy encoding differs from one-hot encoding because it:

- ☒ a. Drops one category to avoid multicollinearity ✓
- ☐ b. Uses frequency information
- ☐ c. Uses -1 instead of 0
- ☐ d. Preserves order

The correct answer is: Drops one category to avoid multicollinearity

**Question 10**

Correct

Mark 0.50 out of 0.50

Generalization refers to a model's ability to perform well on unseen data.

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

**Question 11**

Correct

Mark 0.50 out of 0.50

Overfitting results in high training error and high test error.

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

**Question 12**

Correct

Mark 0.50 out of 0.50

Support vectors are the data points that are:

- ☒ a. Closest to the hyperplane ✓
- ☐ b. Farthest from the hyperplane
- ☐ c. Located at the origin
- ☐ d. Always misclassified

The correct answer is: Closest to the hyperplane

**Question 13**

Correct

Mark 0.50 out of 0.50

Monitoring a deployed model is important because data distributions may change over time.

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

**Question 14**

Correct

Mark 0.50 out of 0.50

A model predicts whether a loan applicant is high risk or low risk, using past labeled data is a type of regression.

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

**Question 15**

Correct

Mark 0.50 out of 0.50

KNN can be used for both classification and regression.

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

**Question 16**

Incorrect

Mark 0.00 out of 0.50

Which encoding is MOST suitable for linear regression with nominal features?

- ☐ a. Ordinal encoding
- ☐ b. Count encoding
- ☒ c. Label encoding ✗
- ☐ d. One-hot encoding

The correct answer is: One-hot encoding

**Question 17**

Incorrect

Mark 0.00 out of 0.50

A model achieves high recall but low precision. What does this indicate?

- ☐ a. Perfect classification
- ☒ b. Few false negatives ✗
- ☐ c. Many false positives
- ☐ d. Few false positives

The correct answer is: Many false positives

**Question 18**

Correct

Mark 0.50 out of 0.50

The margin in SVM is defined as  $\frac{2}{\|w\|}$ .

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

**Question 19**

Incorrect

Mark 0.00 out of 0.50

Email filtering where an email can be tagged as **spam, promotion, and social** at the same time is an example of:

- ☐ a. Binary classification
- ☒ b. Multi-class classification ✗
- ☐ c. Multi-label classification
- ☐ d. Regression

The correct answer is: Multi-label classification

**Question 20**

Correct

Mark 0.50 out of 0.50

Which statement correctly compares linear and logistic regression?

- ☐ a. Both predict continuous outputs
- ☒ b. Logistic regression predicts probabilities ✓
- ☐ c. Logistic regression does not use coefficients
- ☐ d. Linear regression predicts class labels

The correct answer is: Logistic regression predicts probabilities



**Question 21**

Complete

Marked out of 3.00

A binary classifier is evaluated on both the training dataset and the test dataset according to the following confusion matrix.

**Training Confusion Matrix**

|                 | Predicted Positive | Predicted Negative |
|-----------------|--------------------|--------------------|
| Actual Positive | 95                 | 5                  |
| Actual Negative | 3                  | 97                 |

**Testing Confusion Matrix**

|                 | Predicted Positive | Predicted Negative |
|-----------------|--------------------|--------------------|
| Actual Positive | 60                 | 25                 |
| Actual Negative | 15                 | 100                |

- 1- Compute Accuracy and Recall for the test dataset
- 2- Determine whether the model is **overfitting, underfitting, or well-fitted**

For Training confusion Matrix

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN) = 0.96$$

$$\text{Recall} = TP / (TP + FN) = 0.95$$

For Testing confusion Matrix

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN) = 0.8$$

$$\text{Recall} = TP / (TP + FN) = 0.7$$

the model in **overfitting** case because Accuracy rate is 80%

**Question 22**

Correct

Mark 0.50 out of 0.50

Which statement about features and labels is TRUE?

- ☐ a. Labels are created by clustering
- ☒ b. Features are input variables used for prediction ✓
- ☐ c. Labels are optional in supervised learning
- ☐ d. Features are the outputs of the model

The correct answer is: Features are input variables used for prediction